Seyede Marziyeh Mousavi Kani

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Research Interests

- Software Verification
- Compiler Design
- Programming Languages
- Automated Reasoning
- Proof Theory
- Calculus of Computation

Education

B.Sc. in Computer Engineering *2022–2026*

Department of Electrical and Computer Engineering, University of Tehran, Tehran, Iran

Cumulative GPA: 18.39/20

Selected Coursework:

Modeling and Formal Verification of Reactive and Concurrent Systems (18.9/20); Compiler Design and Programming Languages (20/20); Machines and Language Theory (20/20); Algorithm Design (17.75/20)

Diploma in Mathematics and Physics *2016–2022*

National Organization for Development of Exceptional Talents

GPA: 18.8/20

Honors and Awards

- Ranked among the **top 15%** of B.Sc. students in Computer Engineering at the University of Tehran.
- **Ranked 138th, top 0.1%,** among approximately 145,000 participants in the Nationwide Iranian University Entrance Exam in Mathematics and Physics.

Research Experience

TU Wien (Technische Universität Wien) *September 2026–Present*

Predoc Researcher

TU Wien (Technische Universität Wien) *February 2026–September 2026*

Research Intern under the supervision of Dr. Matteo Maffei

Formal Verification of Garbled Circuits

- Formalized garbled circuit constructions and their properties in the Lean theorem prover.
- Developed machine-checked proofs to verify properties of garbled circuit protocols.
- Used Lean to mechanize cryptographic definitions and reasoning about the correctness of garbled circuit constructions.

SoSy Lab (Ludwig Maximilian University of Munich) *March 2025–September 2026*

Under the supervision of Prof. Lee

Extending MoxiChecker

- Implemented interpolation-based algorithms, including IMC and ISMC, to improve performance and strengthen verification capabilities.
- Worked to improve the efficiency of interpolant extraction.

- Improved the reliability and efficiency of verification within the framework.
- Added Bitwuzla and cvc5 support to the model checker.

Formal Methods and Verification Lab (University of Tehran) *May 2025–August 2026*
Under the supervision of Prof. Ghassemi and Prof. Sirjani

Tinification of Rebeca Models

- Modified Rebeca code to generate equivalent TinyTwin representations.
- Applied static analysis techniques to study and reason about system behavior.
- Designed a dependency graph to capture component interactions.

Tehran Institute for Advanced Studies *Summer 2025*
Under the supervision of Prof. Hojjat

SAT-Solver-Based Approach for DFA Decomposition

- Applied SAT solvers to identify decomposed Deterministic Finite Automata (DFAs).
- Extended the approach to determine whether a given DFA can be decomposed into k parts.
- Enabled systematic exploration of possible DFA decompositions.
- Improved the efficiency of analyzing DFA structures.

Projects

CPY Compiler

- Designed and implemented a compiler for CPY, a C-like language with Python-inspired indentation-based scoping.
- Developed key static analysis components, including symbol table construction and type checking.
- Implemented memory management error detection, including use-after-free scenarios.
- Implemented reachability analysis to support optimization by eliminating unreachable code.
- Strengthened expertise in compiler design, language parsing, and optimization techniques through a compiler course final project.

DFA Decomposition

- Implemented SAT-based algorithms for learning and decomposing deterministic finite automata from labeled examples.
- Developed a Python toolchain to encode DFA learning as SAT, reconstruct decomposed DFAs, and validate them against accepting and rejecting string sets.
- Built utilities to visualize decomposed DFAs using DOT/Graphviz and wrote automated tests to ensure language equivalence with the monolithic DFA.

Coursera Automated Reasoning Course

- Gained practical experience with resolution, DPLL, CDCL, Tseitin transformation, and the simplex method.
- Broadened understanding of the heuristics and algorithms that power modern SAT and SMT solvers.
- Strengthened the ability to reason formally about problems and apply automated tools for logic-based verification and optimization.

Coursera Symbolic Model Checking Course

- Applied formal methods to classic verification problems, reinforcing knowledge of temporal logic, state-space exploration, and symbolic representations.
- Strengthened skills in formal verification techniques by using SMT solvers to ensure correctness and consistency in complex systems.

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- Gained hands-on experience in kernel development, low-level programming, and system-level design.

Teaching Assistant Experience

Formal Language and Automata Theory (**Dr. Hojjat**) *Fall 2024, Spring 2025, Fall 2025, Spring 2026*

- Designed questions on PDA (Pushdown Automata), CFG (Context-Free Grammar), CFL, DCFL, reductions, decidability, and recognizability, and graded homework.
- Served as Head Teaching Assistant until September 2026.

Theory of Formal Language and Automata (**Dr. M. Dolati**) *Fall 2024, Spring 2025*

- Designed questions on DFA and NFA, and graded homework.

Programming Languages and Compiler Design (**Dr. F. Ghassemi**) *Fall 2025, Spring 2026*

- Designed course projects and served as Head Teaching Assistant until September 2026.

Database Design Principles (**Dr. A. Shakeri**) *Fall 2025*

- Designed questions and graded homework.

Basics of Data Science (**Dr. M. Khoshnevisan**) *Summer 2024*

- Assisted with course-related academic support, including student guidance and grading-related tasks.

Introduction to Computer Systems and Programming (**Dr. M. Hashemi**, **Dr. M. Moradi**) *Fall 2023*

- Assisted students with introductory programming and computer systems concepts, and supported course-related academic activities.

Volunteer Experience

ACM Mentorship
University of Tehran ACM Student Chapter

Mentored students in an Artificial Intelligence course; designed assignments and projects, supported grading and feedback, and held office hours to clarify core AI concepts and improve student outcomes.

Class Principal
Department of Computer Engineering, University of Tehran

Elected class principal for two consecutive years; acted as a liaison between students and faculty, communicated concerns to the dean and professors, and helped coordinate class and exam schedules to ensure smooth course operations.

Skills

- **Language Skills:**
  - **English:** C1 / IELTS: 8.0
  - **Persian:** Native proficiency

- **Technical Skills:**
  - Compiler development, operating system development, static analysis, model checking, formal verification, hardware design, and simulation
- **Programming Skills:**
  - **Programming Languages:** C, C++, Python, Java, Verilog
  - **Familiar With:** Bash, mCRL2, Lingua Franca, Rebeca, Coq, Lean
  - **Tools:** ANTLR, Z3, Multisim, ModelSim, Vivado, Quartus, Linux, LaTeX